

LEGO® Serious Play® in HRI research: results of a pilot imagining robotic care

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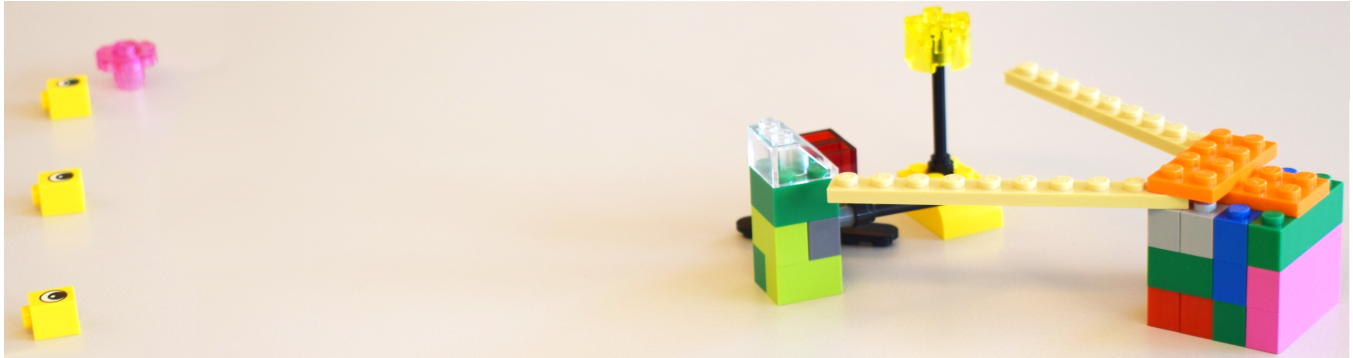


Figure 1: A participant's model of robots for care: single purpose robots (left) impatiently waiting for a nurse at their station

ABSTRACT

In the last ten years, there have been great efforts to increase automation in the health-social care ecosystem, including the use of robotics to provide practical and social care. However, development of these robots often does not include potential users until late in the design process, so they may not adequately address user expectations or needs. This pilot introduces LEGO Serious Play workshops as design tools to support individuals' articulation of the potential benefits and consequences of robot care systems. The narratives elicited address key themes in robotics for care, indicating the workshops' potential use early in design.

CCS CONCEPTS

• **Human-centered computing** → **User centered design.**

KEYWORDS

health-social care, human-robot interaction, LEGO Serious Play

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1 INTRODUCTION

A wide range of studies of human-robot interaction (HRI) in health-social care rely on introducing final-development products into care contexts (e.g., elder care facilities) and observing interaction responses [1, 4] or measuring attitude change [2, 18]. Research earlier in the design process may use videos, photos of robots or written scenarios to gain a more narrative, situated understanding of needs [32]; yet, such studies may still maintain user passivity in the process (e.g., [33]) and have been criticised as limiting respondents to thinking only within those predetermined categories, potentially creating incomplete or misleading outcomes [8]. Collectively, HRI for care hints at a complex relationship between a robot's form, the tasks it performs, and the wider context in which this takes place [25, 37]; however, HRI studies can leave opaque the personal experiences, values and expectations that inform user response [30].

This lacuna urgently needs addressing, particularly as deployment of robots for delivery of care services to physically vulnerable populations is increasingly touted in policy as the remedy for funding shortfalls and workforce shortages [11, 22, 26]. Given the extensive range of robots trialled in care homes which have yet to reach the market years later [29], it is worth considering 1) whether user-centred principles and processes are being usefully applied; 2) whether the values, needs and expectations of care-receivers (and other stakeholders such as care commissioners, professional carers, etc.) are properly understood; and 3) how the early design process could be improved. A range of papers argue for earlier inclusion

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of prospective users in the design process to better identify suitable prototypes before costs and time designing have accumulated [3, 4, 20, 31] but the controlled encounters, thought experiments and questionnaires typically employed do not generally seek to uncover what the respondents previously imagined as 'robot' or 'care' before walking through the door, let alone 'robot for care'. Moreover, methods to research this complexity are few.

We present a pilot workshop using LEGO® Serious Play® (LSP) to investigate tacit 'sociotechnical imaginaries' - projections of a positive, collectively achieved future enabled by development of a specific technology [19] - in the use of robots for health-social care. The intent of the pilot was to a) make visible the sociotechnical imaginaries respondents were tacitly applying when asked about robots for care, b) elucidate what might challenge the positivity (or negativity) of the scenarios imagined and c) evaluate whether LSP can be a productive method for eliciting a useful response to a) and b) in a much larger study with a wide range of stakeholders across the health-social care ecosystem. The full study, *Imagining Robotic Care: Identifying conflict and confluence in stakeholder imaginaries of autonomous care systems*, is now underway.

1.1 LEGO® Serious Play® as a tool for surfacing knowledge and values in design contexts

LSP has its roots in corporate strategising [23], but has also been used to great effect in education [5]. However, its use in HRI has so far been limited to a handful of niche studies [34–36]. LSP deploys a constructionist approach [28], which aims to provide deeper engagement with the problem at hand by simultaneously promoting both individual reflection and knowledge contribution from all members of the group [9, 17]. LSP follows a four-step method: 1) A question is posed; 2) a model is built to support a narrative response to the question; 3) each person shares their story with the group; 4) the group builds on the knowledge uncovered through facilitator-guided discussion [10].

The questions and activities in an LSP workshop are always bespoke, shaped towards the workshop's purpose, goals and length. Relevant to its strengths and weaknesses as a methodology for HRI, LSP has been used in enhancing user involvement in participatory design [12, 16], in design thinking [21, 27] and in aiding a whole-service approach to service change [15]. For our purpose, we were particularly interested in the scenario-creation properties [13] enabled through use of the unusual bricks included in LSP bricksets¹ which can function as metaphors [24] to provide a depth lacking in controlled experiments or investigator-derived scenario research.

2 METHOD

The aims for the pilot were twofold: 1) To determine the suitability of the LSP protocol designed for an HRI context; namely, whether this was broad enough to stimulate creativity yet clear enough to elicit a useful response; and 2) to establish whether the narrative response style typical of LSP would yield data rich enough for a thematic analysis that could inform directions for socially assistive robotic research and development.

2.1 Participants

Participants were recruited via the University of Sheffield's staff and students research volunteer mailing lists. We invited individuals age between 18-30 to attend one of two LSP workshops. Of the 50 respondents, sixteen were invited on a first-come basis, with slight adjustments to ensure an even balance across genders and across native/non-native English speakers. Of those invited to participate, eleven attended the workshops (5 women and 6 men; eight non-native and 3 native speakers). Overall, although it was stressed that no foreknowledge was required, activities showed that both groups were familiar with, and generally positive towards, robotics.

2.2 Ethics

The study received ethical approval from The University of Sheffield before recruitment and informed consent was obtained before workshops began. In workshops, the certified LSP facilitator introduced the 'rules' for participating: the study is a shared and safe space; the meaning of the model belongs to its builder; each person deserves the respect of everyone's attention as they share; the models and their stories, rather than the individuals behind them, are the focus of discussion.

2.3 Materials and Research Environment

The in-person workshops lasted two hours and were video and audio recorded for transcription. The models were also photographed as visual aids to the analysis. Participants worked together at a large, shared table that offered space to construct their LEGO models and share these with the rest of the group. As opposed to commercial LEGO kits, the pieces within LSP kits are selected to discourage concrete representation so individuals may imbue each piece with its own metaphoric meaning (i.e. a round yellow piece may signify sunshine, riches, hope...).

2.4 Protocol

An LSP workshop begins with a series of warm-up exercises designed to familiarise participants with using the bricks as metaphors and building a model to serve as the basis of a story (rather than a literal representation of a robot). As the last step, participants were asked to build a model to help tell a story about a robot they already knew from media (Q1). This question served both to help them understand how to root their scenario in a larger context, and as a baseline to understand their familiarity with science fiction tropes, and how these might influence their subsequent responses.

The Core Questions of the workshop focused on the contextual nature of both 'robot' and 'care' (See Table 1). Participants were first asked to build a model of a story about a *robot providing care to someone* (Q2-care scenario). Both 'care' and the potential 'someone' were left deliberately undefined in order to explore the types of scenarios that immediately sprang to mind; their context, the robot's roles / actions, who was receiving care, and the relationship between the two. The follow-up (Q3-reversal) asked participants to change their model to *reverse the care story's tone* (i.e., complicating positive scenarios or improving uncomfortable ones). This aimed to ascertain the key issue for participants, and whether this was a matter of design, regulation, context, etc.

¹<https://www.lego.com/en-gb/product/starter-kit-2000414> <https://www.lego.com/en-gb/product/window-exploration-bag-2000409>

2.5 Data Analysis

The workshop discussions were anonymised, transcribed, and then analysed using the software package Nvivo 11. Using a grounded theory approach [7], two authors individually open-coded the explanations of the models and subsequent guided discussions into first-order codes, which were then grouped into second-order themes. This coding structure was discussed and refined until authors reached full agreement, resulting in five aggregate themes. Given the small sample size and purpose of the study as a pilot, we did not aim for categorical saturation, but rather sought to explore the themes for their potential for deriving rich data in HRI contexts.

3 RESULTS

Overall, participants imagined a wide variety of robots performing a very wide variety of functions that they saw as embodying aspects of 'care', from the more expected (such as helping nurses in a care home or assisting the elderly with basic daily tasks) to the unexpected (such as helping the user select hairstyle, makeup and outfit to correspond with the activities of the day, or a robot the user could program to destroy everything in a controlled environment as a way of relieving stress). In general, the relationship between robot and human in Q2 was envisioned as reciprocal, ongoing and supportive, with the robot having some semi-autonomous capacity to determine the best way to meet the user's needs. While this pilot study is too small to generalise broadly, five aggregate themes clearly emerged from the Q2 and Q3 scenarios and ensuing group discussions. These aspects of Trust, Comfort, Necessity, Dependence and Control were often interlinked, so that the changes made to the models in Q3 frequently impacted across multiple themes.

3.1 Trust

While trust in HRI has historically defined technically (i.e., that the machine will perform safely and reliably [6, 14]), our respondents built scenarios that interpreted trust as an *interpersonal* quality. Several participants noted that a robot would not be judgmental and therefore could be trusted with the user's weaknesses and vulnerabilities, such as allowing a cleaning robot to see their very messy house, or unburdening oneself of embarrassing or sensitive problems. There was a general agreement that it would be '*easier to ask the robot to be confidential about your information than to ask someone to keep the confidence*' (Person5.Q2). This idea appeared in both one-to-one Q2 care scenarios and in multiple user capacities. In some scenarios, 'care' included a therapeutic function which was expected to increase the humans' confidence and well-being. For example, one participant envisioned a robot tasked with the mental and physical care of company employees, because '*in the working environment people barely share their emotions, barely share their problems*' (Person6.Q2). However, when asked what might change the scenario to its reverse (Q3), it was acknowledged that even the most trusted robot cannot really be trusted, because it would always be vulnerable to interference from other humans:

Person5.Q3 – [the robot] will realise everything...So maybe someone will take advantage of this kind of knowledge, this kind of information about the staff, like the manager, to make a bad influence on this

person. So I think this is a serious problem of my robot.

In other words, robots retaining sensitive data to fulfill their support functions could potentially negate their supportive capacities if end users do not trust the provider of the robot (such as the worker's company, or a local council).

3.2 Comfort

Descriptions of comfort felt by the care recipient in the *robots providing care* (Q2) scenario usually related to the trust in the robot, and the two were often difficult to separate. Nonetheless, several participants created scenarios in which they saw the robot as being a *comforting* presence, especially for the socially isolated, (e.g., providing entertainment functions, facilitating communication with distant family and friends). One respondent's scenario showed the robot as a constant companion, capable of deep, empathic care:

Person1.Q2 – This person without a head is a boy from a divorced family and he was in a totally new country where he doesn't know a person and he doesn't speak the language well so he feels very lonely. And he's a bit confused about life. And this is an old robot...it's going to dead but he's very caring, he's talked with this little boy and he's giving hugs, you see it's a hugging posture...And he can also take the little boy flying, he can caress him and talk to him and make him as confident as other boys to explore things and live life fully.

However, there was some disagreement whether these more companionate robots would need a 'personality' to facilitate the user's comfort with being cared for by a machine, as the following exchange shows:

Person5.Q2 – ...I think it's important for them, for robots to have a personality if they're used for care. Because I think that's a massive part of giving care, is the personality...

Person6.Q2 – I think it's a mistake to give robots personalities. Because then there's nothing to differentiate them from humans... I think it would be really dangerous...

Person4.Q2 – But I think if robots are not designed to have a personality, if they're not programmed for that, people still tend to put personality on them, just see something. Like for example with cars...

Person1.Q2 – There's an emotional attachment.

Taken together, these statements reflect three key positions: one that 'care' requires something beyond a mere machine, the second that it is important for them to remain distinguishably not-human, while the third considers 'personality' as something emerging through user interaction, independent of the developer's choice whether or not to programme the machine with a personable response.

3.3 Necessity

As *comfort* overlapped with *trust*, it also overlapped with themes on how necessary robots were to people's lives, and whether this

Table 1. Protocol for Pilot LSP study

Task	Purpose	Build	Challenge question
Skills building	Familiarity with the bricks	1 min	Build a tower
	Using bricks as metaphor	3 min	Build anything, then choose a term from this card [of concepts such as responsibility, success, etc.] and tell us how your model represents that.
	Narrative construction: Q1 - How has media shaped their concept of 'robot'?	5 min	Build a model to tell a story about a robot you know of. If you can't think of one, make one up.
Core questions	Q2 - What kind of scenario do they imagine? How do they represent 'care'?	5-7 min	Build a model that helps you tell a story about robot giving care to someone. Maybe that person is you, maybe not. Maybe it's a happy story, maybe not.
	Q3 - What determines positive or negative aspects of the story?	5-7 min	Modify the model so that it tells the opposite story. If the person felt comfortable with the robot, what would make them uncomfortable and vice versa?
	Q4 - What do they see as the key element?	1 min	Think about how you feel now about using robots for care. Do you think it's a good idea, a bad idea, an idea that might work depending upon upon context? Then using a red flag, identify the part of your model that best represents that aspect.
Validation	Q5 - Has exploring the other side of the story changed their overall perception?	1 min	Using the [positive and negative] plates, place that part of the model [from Q4] according to whether it makes you feel more positive or more negative about the general idea of robots for care (with the middle being the place for 'it would depend on how that's handled'). Tell us what it is and why you placed it where you did.
	Q6 - Confirming key elements	5 min	Identify your part in a short memorable phrase, something that summarises its importance.

would affect opinion or comfort over time. In some cases this could exacerbate isolation: *'I don't need any more people to help me, I can stay happily alone forever'* (Person5.Q3). For example, one respondent, envisioned a robot that could do housework and be sent to the pub on behalf of its owner so they may meet friends via telepresence. However, in the reversal they noted:

Person4.Q3 – My human got really bored just observing how the robot lives [the human's] life. So first he started doing the ironing, hoovering etc. And then he started going out with friends, going on holidays with them... So the novelty of the robot wore off, and the human just got back into normal life.

Another respondent created a Q2 care scenario where several robots worked alongside a nurse on a hospital ward (Fig 1), but from their perspective the nurse was seen as inefficient and in the way. In the Q3 reversal, the nurse's caring functions and control over the robots was increased, yet the respondent saw deeper problems pertaining to the necessity of robotics in this context:

Person1.Q3 – But then you start going down the path of let's free up the nurses to do the compassion side more by having something that monitors and checks things and does this job, and then we can do this

job, and then you end up to the point where actually is there any reason to have 12 nurses on a ward any more, all running round ...just doing that kind of [compassionate] care? Well, actually let's just have one that runs round while these [robots] do all the practical things. I think there's that thing in manufacturing where you want to increase efficiency and things like that, but in care is there really that kind of desperate need for it or not?

Here the scenario aspect of LSP allowed the respondent to move beyond the usual reasons given for automation in hospital or care home settings to consider the broader implications of wards filled with many robots, challenging the assumption that automating the physical work of care brings unquestionable benefits.

3.4 Dependence

Dependence on robots was closely tied to necessity. Several respondents, particularly the younger men, modelled a robot who would cook and clean and get them out of bed in the morning, something 'you can program like a friend, or intimacy, or housekeeping or parents' (Person2.Q2). Rather than a wife or girlfriend however, in this clearly gendered scenario the robot seemed to be imagined as a surrogate, idealised mother. However, the same respondents did

agree in the Q3 reversal that such robots could promote a state of laziness leading to actual incapacity to care for one's self, particularly when used by those who have no need to depend on robots to carry out these functions:

Person2.Q2 – ...I think it's not very healthy. But on the other hand, we're expecting some superior service from the robot machines. In the future, not only the jobs but also the emotional part of the human brain could be damaged by this kind of thing.

Another participant in the same group noted that:

Person6.Q3 – ... when the robot breaks or just isn't doing the right thing that it's supposed to be doing, it isn't fulfilling its role of caring and the whole support system just crumbles because the people are relying on it. So as soon as something goes wrong, it's all bad stuff. So it all just kind of falls down.

Here again we see LSP's capacity to encourage participants to think beyond one particular scenario, and explore how the use of robotics, in and of itself, could encourage a potentially dangerous systemic dependence, critical in a field in which – because of the nature of care activities – breakdown could become life-threatening.

3.5 Control

Finally, the extent to which the robot could be trusted and depended upon (in a positive sense) was seen as critically mediated by how (and by whom) control of the robot could be maintained. In these scenarios the main worry was that other humans, by accessing either the data or the programming, could gain a level of control not only over the robot, but over the user's life – potentially literally: *'it's so easy to hack in, so if people want to kill you they just program this kind of a code and you'll be dead.'* (Person2.Q3). Loss of control due to the robots' operating parameters was also a concern; in the hospital narrative, for example, the robots' efficiency was compromised, potentially forcing the nurse to adapt her behavior to their needs, rather than the other way around:

Person3.Q3: ...perhaps the robots are doing things like the heavy lifting of patients and monitoring them and cleaning the wards and things like that. Leaving the nurse to just care for people and talk to them. ... But these [robots] are all over here in the staff room complaining that they've basically been designed to be as efficient as possible but the nurse keeps getting in the way of them being able to do their job by trying to talk to patients and care for them.

In general, the aspect of the Q2 care scenario model which changed when the tone was reversed in Q3 had more to do with human control than robotic design. As the 'killer robot' respondent noted, *'it's not the robot that changed, it's the function of the robot ... it's people who use the knife who kill people'* (Person2.Q3). In the Q3 model for the warm-hearted robot scenario, for example, the boy's control was lost to public resistance to robots in society, so that he became violent in defending his fatherly robot (Person3.Q3).

Others in the care scenario reversal (Q3) discussions spoke of taking back control by changing *'the human position, from being*

completely lazy to starting to do something. So [changing] the attitudes, how the technology can be used' (Person4.Q3) or *'not just leaving [robots] to their own devices but having people behind the scenes so it's not just completely lacking humanity'* (Person6.Q3), considering not just the robot itself, but how it should be embedded in existing systems.

Overall, there appeared to be more confidence in the robot's ability to perform its functions reliably than in the human's ability to design robots in a way that would be truly beneficial and safe from other human's malevolent desires. The complexity of the narratives suggest 'control' can function across many (and potentially all) of the relational contexts – between robot and user, provider and user, robot and society – and include both intentional and unintentional results, the latter being a perhaps neglected aspect of conceptualising machine-aided care. While the nurse scenario was the clearest example, most of the stories indicated that processual care and empathic care, while difficult to disentangle, may need to be approached as separate domains. The workshops therefore surfaced conflicting imaginaries of 'care'.

4 DISCUSSION

Our pilot served two purposes: 1) to test the suitability of the protocol questions, and 2) to determine the efficacy of LSP as a methodology for eliciting a rich, contextualised understanding of 'robots' and 'care' by not pre-determining scenarios or definitions. The transcribed narratives indicated that the Core Questions were effective in eliciting responses that were at once deeply personal, yet situated in a complex sociotechnical imaginary which went beyond the usual scenarios envisioned in HRI research, and helped to make more visible the activities, people, and relationships that might be displaced by automated care. These were elaborated in the five aggregate dimensions (Trust, Comfort, Necessity, Dependence and Control) which encapsulated the key elements upon which benefit and detriment turned (see Table 2). Across these themes, we also found a strong emphasis on interaction between robot and user as key to feeling cared-for, countering the tendency of developers to imagine potential users as passive recipients [4], rather than active participants in care. We also found the thematic richness was reinforced by expanding the stories from the original care scenario (Q2) to explore how it might look in reverse (Q3), rather than moving to a new question entirely, as it allowed the respondents' own tacit assumptions to surface and be questioned.

Our pilot confirmed that LSP was a useful approach which could generate strong and relevant themes for analysis without requiring prompt materials such as photos or videos of robots, central to other design studies [3, 20, 31]). It allowed 'care' to be defined in ways (e.g. as something that can be trusted with secrets, alleviate some of the day-to-day tasks of adult life, or help to navigate a confusing social world) which show themselves as potentially not unique to users of social care services, but certainly of relevance to that group. The narratives confirmed that respondents could imagine robots providing the emotionally supportive functions associated with 'care', but also helped to identify potential tipping points, exploring the balance between technological and emotional factors which will be necessary if robots are to have their intended benefits in this field.

Table 2. Themes Arising from the Stories

Trust	A robot does not judge, may increase confidence to discuss delicate issues with a machine instead of a human	Security and connectivity are worries, the robot could access other data about you, or could be hacked to reveal your secrets
Comfort	Companionship, especially for the isolated	May impede connection with other humans
Necessity	Unpleasant or hard tasks could be automated	Robots may be an imposition at times
Dependence	Once we have them, how dependent will we become on them and for what kinds of things?	Yet another social division, they become something like an addiction
Control	Of the programming, of who has them or access to them, of whose needs are really served	At what point do the robots (or those behind them) end up controlling us?

As with any pilot-level work, the results discussed herein are more indicative of the quality of the methodological innovation than a substantive statement on the topic of study. The small sample of participants means that there was no expectation that all potential themes, or even all aspects of the current themes, have been identified. The population considered in this study may also hold specific and shared imaginaries owing to their age, backgrounds and current circumstances as members of a university. Further research to substantiate both LSP as a method for facilitated focus group discussion and the qualitative value of sociotechnical imaginaries as an analytic concept will be needed to firmly establish their utility in HRI.

To this end, we are presently using a slightly refined version of the protocol presented here (keeping Q2 and Q3 intact) with a range of key stakeholder groups in the health-social care ecosystem (including, but not limited to: roboticists and designers, care commissioners, social care providers, social workers, academics in social care and in HRI, paid and unpaid/family carers, social care users and general publics from 18-80.) Identifying where these imaginaries cohere and conflict, and how to incorporate this diversity of stakeholder input as part of problem definition (before robot solutions are envisioned) is expected to contribute to discussions of user-centred design, as well as help illuminate where existing research trajectories in care robotics may need refinement before costly mistakes in development occur by providing a clearer picture of the context(s) into which they are likely to be deployed.

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